

SRD detailed balance does not imply GNS detailed balance

A primitive qubit Lindblad counterexample

Source target: arXiv:1810.00906, Remark IV.5(2)

Status: candidate full counterexample; likely valid pending human review.

Source question

Yu Cao, Jianfeng Lu, and Yulong Lu, *Gradient flow structure and exponential decay of the sandwiched Rényi divergence for primitive Lindblad equations with GNS-detailed balance*, arXiv:1810.00906; J. Math. Phys. **60** (2019), 052202, prove

GNS detailed balance $\implies$ SRD detailed balance $\implies$ KMS detailed balance.

Remark IV.5(2), on PDF page 23, asks whether the first implication reverses:

It is still unknown whether the SRD detailed balance condition is equivalent to the GNS detailed balance condition or not.

C. Relations between GNS, SRD, KMS detailed balance conditions

Proposition IV.4. *For a primitive Lindblad equation with the unique stationary state $\sigma \in \mathfrak{D}_+$,*

- 1. GNS detailed balance condition implies SRD detailed balance condition.*
- 2. SRD detailed balance condition implies KMS detailed balance condition, while conversely it is generally not true.*

Remark IV.5. 1. This suggests that the class of Lindblad equations with GNS detailed balance considered in Theorem I.2 seems to be general enough, because it is impossible to generalize Theorem I.2 to the class of Lindblad equations with KMS detailed balance by Proposition IV.4. More specifically, as we shall show later in the proof of Proposition IV.4, there exists at least one primitive Lindblad equation that satisfies KMS detailed balance condition, but does not satisfy SRD detailed balance condition (in particular, $\mathcal{W}_{\sigma,\alpha} \circ \mathcal{L} \circ \mathcal{W}_{\sigma,\alpha}^{-1} \neq \mathcal{L}^\dagger$ for any $\alpha \neq 2$); because SRD detailed balance condition is necessary for having a gradient flow structure, this particular Lindblad equation cannot be expressed as the gradient flow dynamics of $D_\alpha(\rho||\sigma)$ for any $\alpha \neq 2$.

2. It is still unknown whether the SRD detailed balance condition is equivalent to the GNS detailed balance condition or not. Thus it might be interesting to characterize Lindblad equations with SRD detailed balance, though this task seems to be rather technical.

Proof of Proposition IV.4. The first statement comes immediately by combining Theorem I.2 and Theorem IV.1. As for the second statement, since the SRD detailed balance condition contains KMS detailed balance condition as a special instance when $\alpha = 2$, obviously SRD implies KMS. In the following, we provide a primitive two-level Lindblad equation with KMS detailed balance from [18, Appendix B] to demonstrate that KMS does not imply SRD.

Let us consider $K_1 = |\psi\rangle\langle 0|$ and $K_2 = |\phi\rangle\langle 1|$ where $|\psi\rangle = \frac{1}{\sqrt{2}}(|0\rangle + |1\rangle)$ and $|\phi\rangle = \frac{1}{\sqrt{5}}(|0\rangle + 2|1\rangle)$.

For a faithful invariant state σ , the source defines SRD detailed balance by requiring the Heisenberg generator $\mathcal{L}$ to be self-adjoint for every inner product

$$\langle A, B \rangle_{\mathcal{W}_{\sigma,\alpha}} = \text{Tr}(A^* \mathcal{W}_{\sigma,\alpha}(B)), \quad \alpha > 0.$$

Equivalently, $\mathcal{W}_{\sigma,\alpha} \mathcal{L} \mathcal{W}_{\sigma,\alpha}^{-1} = \mathcal{L}^\dagger$.

Proof intuition

In a basis diagonalizing a qubit state, every SRD weight gives the *same* weight to the two oppositely oriented coherences E_{01} and E_{10} . GNS does not: it weights them by the two different eigenvalues of the invariant state. We therefore add a Pauli- X noise term that mixes E_{01} and E_{10} . This mixing is symmetric for every SRD geometry, but it is not GNS symmetric when the stationary state is nonracial. Asymmetric up/down jumps make that stationary state nonracial while preserving a primitive GKSL generator.

Counterexample theorem

Let $E_{ij} = |i\rangle\langle j|$ in the standard basis of $\mathbb{C}^2$ and $X = E_{01} + E_{10}$. For $a, b, c > 0$, define

$$V_+ = \sqrt{a} E_{10}, \quad V_- = \sqrt{b} E_{01}, \quad V_x = \sqrt{c} X$$

and the Heisenberg generator

$$\mathcal{L}(A) = \sum_{V \in \{V_+, V_-, V_x\}} \left(V^* A V - \frac{1}{2} \{V^* V, A\} \right). \quad (1)$$

Theorem 1. *If $a, b, c > 0$ and $a \neq b$, then (1) generates a primitive quantum Markov semigroup which satisfies SRD detailed balance for every $\alpha > 0$ but does not satisfy GNS detailed balance. In particular, SRD detailed balance is not equivalent to GNS detailed balance, already on a qubit.*

Proof. Put

$$A_0 = a + c, \quad B_0 = b + c, \quad p_0 = \frac{B_0}{A_0 + B_0}, \quad p_1 = \frac{A_0}{A_0 + B_0}, \quad \sigma = p_0 E_{00} + p_1 E_{11}.$$

The expression (1) is in GKSL form and each summand annihilates the identity, so it generates a completely positive unital semigroup. Direct multiplication gives

$$\mathcal{L}(E_{00}) = -A_0 E_{00} + B_0 E_{11}, \quad \mathcal{L}(E_{11}) = A_0 E_{00} - B_0 E_{11}, \quad (2)$$

$$\mathcal{L}(E_{01}) = -\gamma E_{01} + c E_{10}, \quad \mathcal{L}(E_{10}) = c E_{01} - \gamma E_{10}. \quad (3)$$

where $\gamma = (a + b)/2 + c$. Thus, in the ordered Hilbert–Schmidt orthonormal basis $(E_{00}, E_{11}, E_{01}, E_{10})$,

$$[\mathcal{L}] = \begin{pmatrix} -A_0 & A_0 & 0 & 0 \\ B_0 & -B_0 & 0 & 0 \\ 0 & 0 & -\gamma & c \\ 0 & 0 & c & -\gamma \end{pmatrix}. \quad (4)$$

Since $p_0 A_0 = p_1 B_0$, equation (4) shows that $\mathcal{L}^\dagger(\sigma) = 0$.

We now verify SRD symmetry. The source paper diagonalizes its weight operator in the matrix-unit basis. For every $\alpha > 0$ there is a positive number q_α such that

$$[\mathcal{W}_{\sigma, \alpha}] = \text{diag}(p_0, p_1, q_\alpha, q_\alpha). \quad (5)$$

Indeed, for $\alpha \neq 1$ its two off-diagonal coefficients are both

$$q_\alpha = (\alpha - 1) \frac{p_0^{1/\alpha} - p_1^{1/\alpha}}{p_1^{(1-\alpha)/\alpha} - p_0^{(1-\alpha)/\alpha}},$$

which is symmetric under $p_0 \leftrightarrow p_1$; at $\alpha = 1$ their common value is the logarithmic mean $(p_0 - p_1)/(\log p_0 - \log p_1)$.

Multiplying (4) by (5), one obtains

$$\mathcal{W}_{\sigma, \alpha} \mathcal{L} = \mathcal{L}^\dagger \mathcal{W}_{\sigma, \alpha}.$$

On the diagonal block this is exactly the identity $p_0 A_0 = p_1 B_0$; on the coherence block it follows because the block in (3) is real symmetric and both coordinates have weight q_α . Since $\mathcal{W}_{\sigma,\alpha}$ is invertible, the defining SRD identity holds for every $\alpha > 0$.

For GNS detailed balance, the source uses $\langle A, B \rangle_1 = \text{Tr}(\sigma A^* B)$. Relative to our ordered basis its weight operator is right multiplication by σ , hence

$$[R_\sigma] = \text{diag}(p_0, p_1, p_1, p_0).$$

The coherence-flip entries in (4) would be GNS symmetric only if $cp_1 = cp_0$. More explicitly,

$$\langle \mathcal{L}(E_{01}), E_{10} \rangle_1 = cp_0, \quad \langle E_{01}, \mathcal{L}(E_{10}) \rangle_1 = cp_1.$$

Because $c > 0$ and $a \neq b$ imply $p_0 \neq p_1$, these quantities differ. Thus GNS detailed balance fails.

Finally, the four eigenvalues of (4) are

$$0, \quad -(A_0 + B_0), \quad -\frac{a+b}{2}, \quad -\frac{a+b}{2} - 2c.$$

The zero eigenspace is precisely $\mathbb{C}I$, all other eigenvalues are strictly negative, and σ is faithful. Hence the predual semigroup converges to $\sigma \text{Tr}(\cdot)$ and is primitive. $\square$

Smallest explicit instance

Take $(a, b, c) = (1, 3, 1)$. Then

$$\sigma = \begin{pmatrix} 2/3 & 0 \\ 0 & 1/3 \end{pmatrix}, \quad [\mathcal{L}] = \begin{pmatrix} -2 & 2 & 0 & 0 \\ 4 & -4 & 0 & 0 \\ 0 & 0 & -3 & 1 \\ 0 & 0 & 1 & -3 \end{pmatrix},$$

with spectrum $\{0, -2, -4, -6\}$. This single concrete generator already answers the source question negatively; the three-parameter family shows that the failure is robust rather than isolated.

Verification

The accompanying script constructs the three Lindblad operators, derives the superoperator from (1), checks the invariant state and unitality, tests the SRD intertwining identity at seven Rényi orders, confirms a nonzero GNS residual, and checks the exact predicted spectrum. For the explicit instance it reports maximum sampled SRD residual 3.14×10^{-16} and GNS residual 0.4714045 in Frobenius norm. These computations are supplementary; the proof above is exact.

Novelty check, limitations, and review request

A bounded search on 17 August 2026 used the exact arXiv id, title, DOI, authors, the exact phrase “SRD detailed balance”, “sandwiched Rényi detailed balance”, and combinations with GNS/KMS/equivalence/Lindblad. Local run indexes and web/arXiv-oriented searches found the source paper, the first author’s thesis restating the same open remark, and papers citing the gradient flow result, but no later answer and no occurrence of this qubit family. The novelty assessment is therefore positive but bounded, not exhaustive.

This packet disproves only the proposed equivalence. It does not characterize all SRD-detailed-balanced generators and does not close the broader gap between necessary and sufficient conditions for arbitrary Rényi gradient-flow structures. Human review should focus on the convention relating weighted self-adjointness to $\mathcal{W}\mathcal{L} = \mathcal{L}^\dagger \mathcal{W}$, the orientation of the two GNS weights, and the standard equivalence between the displayed spectral calculation and primitivity.

References

1. Y. Cao, J. Lu, and Y. Lu, *Gradient flow structure and exponential decay of the sandwiched Rényi divergence for primitive Lindblad equations with GNS-detailed balance*, J. Math. Phys. **60** (2019), 052202; arXiv:1810.00906.
2. V. Gorini, A. Kossakowski, and E. C. G. Sudarshan, *Completely positive dynamical semigroups of N -level systems*, J. Math. Phys. **17** (1976), 821–825.
3. G. Lindblad, *On the generators of quantum dynamical semigroups*, Commun. Math. Phys. **48** (1976), 119–130.